ConfCtrl: Enabling Precise Camera Control in VideoDiffusion via Confidence-Aware Interpolation

Liudi Yang, George Eskandar, Fengyi Shen, Mohammad Altillawi, Yang Bai, Chi Zhang, Ziyuan Liu and Abhinav Valada

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Challenge

SPARSE INPUT — 2 images
Wide viewpoint gap and unseen regions

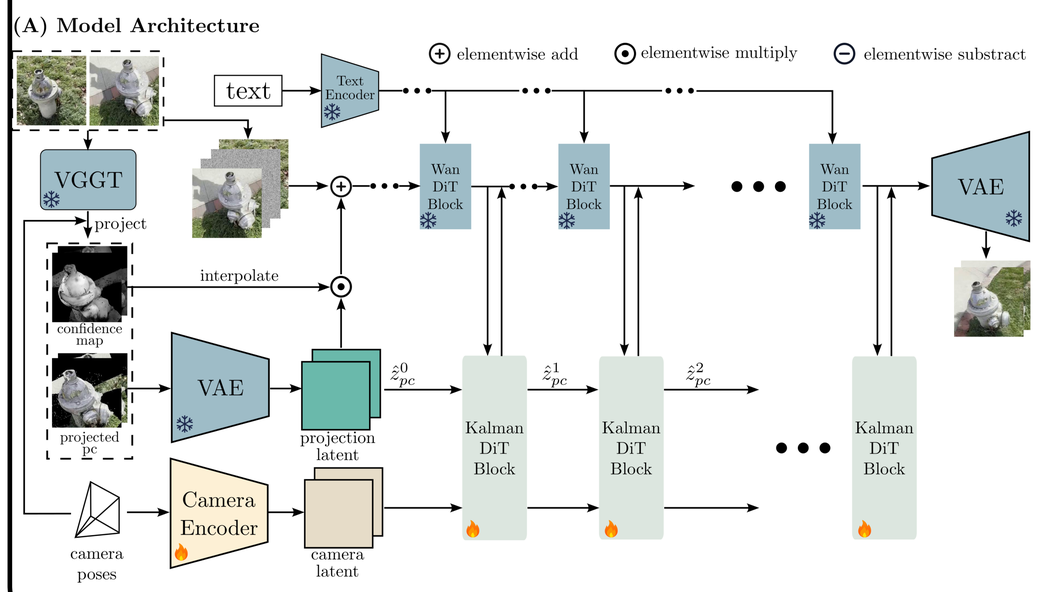
REGRESSION
- Precise Camera Control ✓
- Unseen Completion ✗

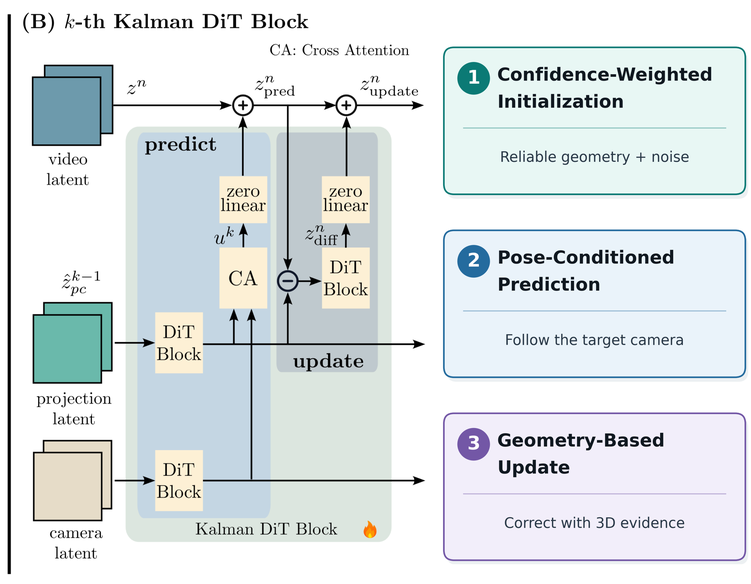
VIDEO DIFFUSION
- Precise Camera Control ✗
- Unseen Completion ✓

NEED BOTH

ConfCtrl — OUR SOLUTION
- Precise Camera Control ✓
- Unseen Completion ✓

Method



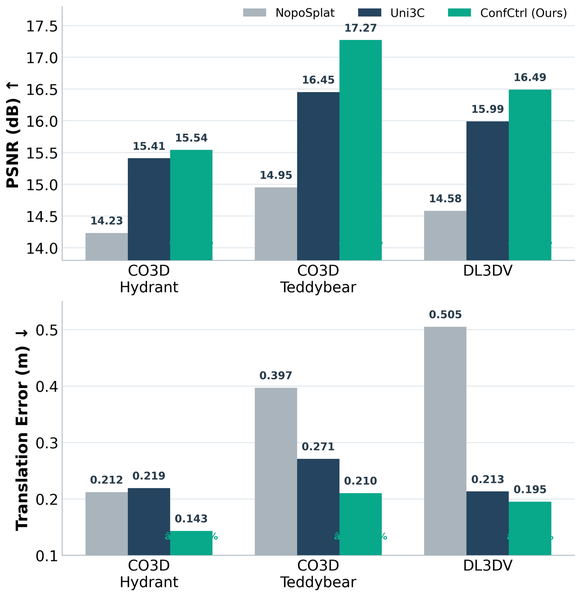


1 Confidence-Weighted Initialization
Reliable geometry + noise

2 Pose-Conditioned Prediction
Follow the target camera

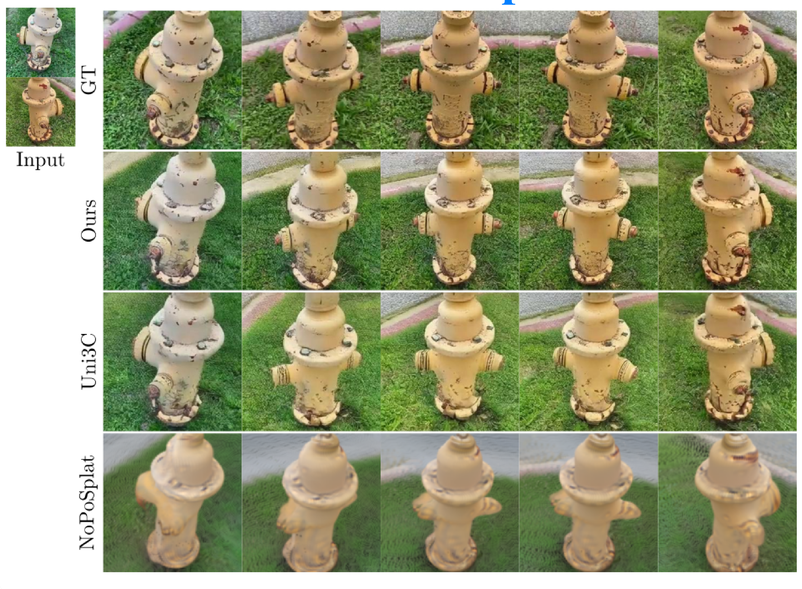
3 Geometry-Based Update
Correct with 3D evidence

Quantitative Results



Qualitative Results

Baseline Comparison



Zero-Shot to New Dataset

